

Goutam Sharma

Software Development Engineer (SDE) | AI/ML Engineer

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PROFESSIONAL SUMMARY

B.Tech Computer Science student with expertise in Data Structures & Algorithms, Software Development, and Artificial Intelligence. Hands-on experience in machine learning research, scalable application development, and AI system design using Python, NLP, computer vision, and Large Language Models (LLMs). Demonstrated ability to build end-to-end solutions through research internships and industry-relevant projects.

EDUCATION

- IES IPS Academy** Indore, India
B.Tech in Computer Science and Engineering — **CGPA: 9.33/10** Aug 2023 – Present
Relevant Coursework: Data Structures & Algorithms, Operating Systems, Machine Learning, DBMS, Computer Networks

SKILLS SUMMARY

- Programming Languages:** C++, Python, SQL
- Computer Science Fundamentals:** Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, Low Level Design
- Machine Learning & AI:** Machine Learning, Deep Learning, Natural Language Processing, Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Computer Vision
- Frameworks & Libraries:** PyTorch, TensorFlow, Scikit-Learn, Hugging Face Transformers, LangChain, LangGraph
- Databases & Deployment:** MySQL, Git, Docker, MLOps
- Soft Skills:** Problem Solving, Analytical Thinking, Team Collaboration, Adaptability, Time Management

EXPERIENCE

- Indian Institute of Technology Gandhinagar** Onsite
Research Intern | Supervisor: Prof. Mayank Singh May 2026 – Present
 - Multimodal Data Pipeline Development:** Designed and developed data pipelines for large-scale image-text datasets, including image extraction, caption generation, data preprocessing, and metadata management for Vision-Language Model (VLM) workflows.
 - Scene Graph VLM Research:** Conducted an in-depth study of recent Scene Graph-based Vision-Language Model papers, focusing on spatial reasoning, visual relationship understanding, and cross-modal representation learning techniques.
 - Dataset Curation & Quality Validation:** Worked on multimodal dataset organization, data cleaning, quality validation, and annotation workflow optimization to support scalable multilingual AI and VLM development.
- Indian Institute of Technology Indore** Remote
Research Intern | Supervisor: Prof. Nagendra Kumar Jan 2026 – Present
 - AI/ML Research:** Exploring deep learning and natural language processing techniques through hands-on experimentation with transformer-based models and modern machine learning frameworks.
 - Machine Learning Pipeline Development:** Building end-to-end ML workflows covering data preprocessing, model training, fine-tuning, evaluation, and performance analysis using PyTorch.
 - Research & Experimentation:** Conducting structured experiments, analyzing results, and maintaining reproducible workflows while studying recent research papers and implementing proposed methodologies.
- Indian Institute of Information Technology Vadodara** Onsite
Research Intern | Supervisor: Prof. Deepika Gupta May 2025 – Jul 2025
 - Multi-Script OCR Development:** Developed a multilingual OCR system supporting six scripts using CNN, ResNet, VGG, and CNN-RNN architectures in PyTorch for character and text recognition tasks.
 - Data Processing & Model Engineering:** Built data preprocessing and training pipelines, addressing script-specific challenges such as ligatures, diacritics, character variations, and bidirectional text across multiple languages.
 - Performance Optimization:** Improved recognition performance to over 99% accuracy with a 30% reduction in Character Error Rate (CER), and further enhanced efficiency using a hybrid quantum-classical CNN that achieved 99.8% accuracy while reducing model parameters by approximately 40%.

PROJECTS

- Neurolytix AI – AutoML Platform | GitHub:** Developed an AutoML platform using Streamlit that automates data exploration, model training, evaluation, and report generation. Implemented multiple machine learning algorithms including XGBoost, LightGBM, and CatBoost, along with hyperparameter tuning, fairness analysis, and time-series forecasting capabilities.
- Brain Tumor Classification using EfficientNetB1 + GradCAM | GitHub:** Built a deep learning pipeline for brain tumor classification from MRI scans using EfficientNetB1 and transfer learning. Applied image preprocessing, data augmentation, and GradCAM-based visual explanations to improve model interpretability and performance analysis.
- Vision-Language Model | GitHub:** Implemented a Vision-Language Model pipeline using CLIP and Qwen2.5-VL for image-text understanding tasks, including image captioning, zero-shot classification, semantic retrieval, and language-guided object localization. Optimized inference workflows for efficient GPU utilization.

CERTIFICATIONS

- Build a Computer Vision App with Azure Cognitive Services – Microsoft | Credential:**
- Deep Learning and Reinforcement Learning – IBM | Credential:**
- Google Certified Professional Machine Learning Engineer – Google | Credential:**